

Linear Congruential Generator

We have designed and implemented a new prime factorization algorithm. It is believed that the most challenging inputs to the algorithm are integers that are the product of K distinct primes. We aim to test the algorithm on a sequence of integers generated by a linear congruential generator. The question is how to select the generator seed to generate many challenging inputs.

The task

You are given the parameters of a linear congruential generator. Your task is to compute the seed of the generator that produces a sequence of N pseudorandom values containing as many integers as possible whose prime factorization consists of exactly K distinct primes.

Input

The input is a single line containing integers A , C , M , K , and N separated by a space. The values A , C , and M define the linear congruential generator given by the formula:

$$x_{i+1} = (A \times x_i + C) \bmod M$$

It is guaranteed that the generator has a period of length M .

Value K is the number of prime factors of the considered most challenging inputs, N specifies the number of inputs to generate from the chosen seed to test the algorithm. The generated values are $x_1, x_2, \dots, x_N$, where x_1 equals the seed.

Values of A , C and M are not greater than 2×10^7 . Moreover, $1 \leq K \leq 10$ and $1 \leq N \leq M$.

Output

The output is a single line containing two integers S and I separated by a space. S is the optimal seed, and I is the number of the most challenging inputs generated from S . If multiple seeds generate the same number of challenging inputs, S is the number that appears first in the generator's sequence when starting with 0.

Example 1

Input

5 11 8 1 4

Output

0 3

The generator produces the sequence: 0, 3, 2, 5, 4, 7, 6, 1. Since $K = 1$, the most challenging inputs are primes. If seed 0 is chosen, primes 2, 3, and 5 are included in the generated sequence of length $N = 4$. This is the optimal setting, as a sequence of length 4 cannot include all 4 primes produced by the generator.

Example 2

Input

5 3 16 2 7

Output

8 3

The generator produces the sequence: 0, 3, 2, 13, 4, 7, 6, 1, 8, 11, 10, 5, 12, 15, 14, 9. Since $K = 2$, the most challenging inputs are numbers in $[0, 15]$ that are products of two distinct primes (6, 10, 14, 15). The optimal seed is 8.

Example 3**Input**

17 9 32 2 10

Output13 5

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.

[Link to public data set](#)